BHARATI VIDYAPEETH'S COLLEGE OF ENGINEERING

(APPROVED BY AICTE, AFFILIATED TO GURU GOBIND SINGH INDRAPRASTHA UNIVERSITY)

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DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

MICROELECTRONICS LABORATORY – ECC-353: CONTENT BEYOND SYLLABUS (CBS)

Academic Year: 2026–2027

EXPERIMENT NO. 11: Laboratory CBS: Implementation and Simulation of the ECC-305 Selected Circuit

STUDENT DETAILS

Name of Student: _____ **University Roll No.:** _____

Batch: _____ **Group No.:** _____

Date of Submission: _____ **Student Signature:** _____

EVALUATOR'S REMARKS

Total	_____ **/ 10**

Evaluator Name: _____

Evaluator Signature: _____

Date: _____

TITLE OF EXPERIMENT

Implementation and Simulation of a Break-Before-Make CMOS Inverter for Low-Power Delay Generation

CIRCUIT SELECTED FROM ECC-305

Circuit Name: Break-Before-Make (BBM) CMOS inverter / delay cell (PtNf full-swing topology, Fig. 9(b) and Fig. 8 of the paper), compared with a conventional 4-inverter CMOS delay cell.

Low-Power Technique: Break-before-make (dual-ramp, time-skewed) switching to eliminate direct-path (short-circuit) current; supply charge/energy per delay is reduced.

1. AIM

To implement and simulate the selected CMOS circuit using the low-power design technique studied during the Microelectronics Theory (ECC-305) research activities and to evaluate its performance.

2. OBJECTIVES

- To understand the transistor-level architecture.
- To implement the selected CMOS circuit.
- To verify its functional operation.
- To simulate the circuit using the prescribed tool.
- To measure power and delay.
- To calculate PDP.
- To compare the reference and optimized circuit.
- To evaluate the effectiveness of the selected/implemented low-power technique(s).

3. RESEARCH REFERENCE

Research Paper Title: Break-before-Make CMOS Inverter for Power-Efficient Delay Implementation

Author(s): J. Puhan, D. Raič, T. Tuma, and Á. Bűrmen

Journal/Conference: The Scientific World Journal, vol. 2014, Article ID 349131 (doi:10.1155/2014/349131)

Year: 2014

4. SOFTWARE / SIMULATION ENVIRONMENT

Simulation Tool: ngspice-42 (open-source SPICE, batch mode, Ubuntu 24 container). Transient analysis; Python 3 / NumPy used only for post-processing and cross-check. *Not* the paper's HSPICE.

Technology Node / Model: *Substitute* Level-1 (Shichman–Hodges) 0.18 µm-class model built from the parameters in the caption of Fig. 4 of the paper ($V_{TN} = 0.5\,\text{V}$, $V_{TP} = -0.5\,\text{V}$, $K_{PN} = 350\,\mu\text{A/V}^2$, $K_{PP} = 90\,\mu\text{A/V}^2$, $C_{ox} = 8.6\,\text{fF/}\mu\text{m}^2$, $L_{\min} = 180\,\text{nm}$). It is *not* the TSMC 0.18 µm HSPICE model used by the authors (proprietary); λ, γ, ϕ, junction and overlap capacitances are my assumptions (file `models_180nm_L1.lib`).

V_{DD}**:** 1.8 V (same for both circuits)

Temperature: 27 °C (ngspice `.temp 27`; Level-1 parameters are not temperature-tuned)

Load Capacitance: $C_L = 2\,\text{fF}$ at the output of the output unit inverter ($\approx$ one unit-inverter input capacitance); a sweep over 0.5–10 fF was also run. Input: 0–1.8 V pulse, 50 ps edges, 20 ns period (50 MHz), 50 % duty.

5. CIRCUIT DIAGRAM

Transistor-level schematics of the two circuits exactly as simulated (redrawn from paper Figs. 8 and 9; sizes W/L in nm).

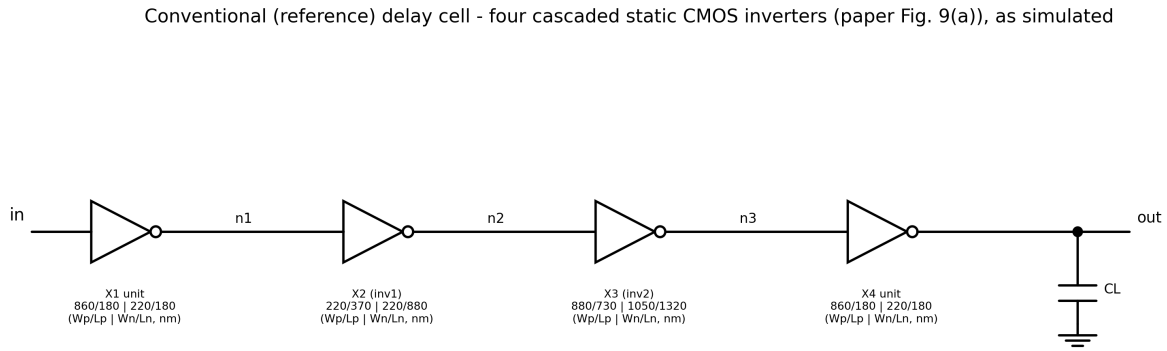


Figure 1. Reference: conventional delay cell (4 static CMOS inverters). Netlist: `conventional_inverter.sp`.

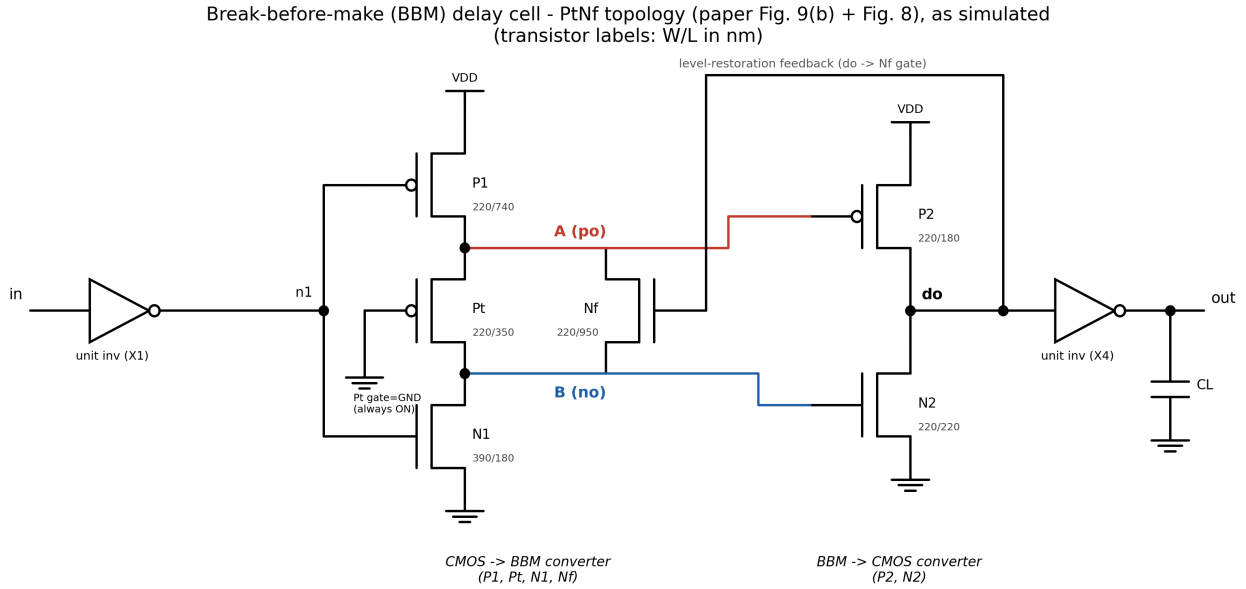


Figure 2. Optimised circuit: BBM delay cell, PtNf topology. Netlist: `bbm_inverter.sp` (sizes shown are the paper's 750 ps sizing).

6. DESIGN / IMPLEMENTATION DETAILS

Circuit reconstruction. The paper's Fig. 9(b) delay cell is: input unit inverter → split-output BBM inverter (CMOS-to-BBM converter, Fig. 1(a)) → split-input merge inverter (BBM-to-CMOS converter, Fig. 1(c)) → output unit inverter. The bidirectional delay element is the PtNf full-swing implementation of Fig. 8: P_1 (source V_{DD}), node A (p_o), P_t (PMOS, gate tied to GND, i.e. always conducting in triode), node B (n_o), N_1 (source GND); N_f is an NMOS connected across A–B whose gate is driven by the next-stage output (d_o) to restore the low level of A. Stage 3 has P_2 (gate A) and N_2 (gate B) with joined drains (d_o). These connections were read directly from the figures in the PDF; the polarity of N_f's gate (connected to the merged output) and the two-transistor stage 3 follow the paper's statement that the level-restoration control signals are taken from the next-stage output. Transistor count: conventional 8; BBM 10 (two unit inverters = 4, plus $P_1, P_t, N_1, N_f = 4$, plus $P_2, N_2 = 2$).

Sizing used. Both cells use the sizing printed in the Fig. 11 caption of the paper for $T_p = 750\,\text{ps}$. The unit inverters are not specified by the paper; I chose $W_p/W_n = 860\,\text{nm}/220\,\text{nm}$, $L = 180\,\text{nm}$, following the symmetric stage of Fig. 4. Conventional: inv1 $W_p/L_p = 220/370$, $W_n/L_n = 220/880$; inv2 $W_p/L_p = 880/730$,

$W_n/L_n = 1050/1320$ (nm). BBM: P_1 220/740, N_1 390/180, P_t 220/350, N_f 220/950, P_2 220/180, N_2 220/220 (nm). Source/drain parasitics follow the paper's rule $A_d = A_s = 0.8\,\mu\text{m} \times w$, $P_d = P_s = 1.5\,\mu\text{m} + w$.

Measurement definitions (identical for both circuits). Delay: 50 % (0.9 V) crossing of the input to 50 % crossing of the cell output, rising and falling, third edge of each (steady state), $T_p = (t_{pLH} + t_{pHL})/2$. Charge: supply charge of the whole cell (all four stages; the input source is not counted) integrated over 4 complete cycles (21–101 ns) and divided by 4, giving Q per cycle (one rising plus one falling slope, the paper's metric). Energy per cycle $E = QV_{DD}$; average power $P_{avg} = E/T_{cycle}$ at 50 MHz; PDP $= P_{avg} \times T_p$. Because E per cycle is independent of frequency (leakage is zero in a Level-1 model), P_{avg} and PDP scale with the chosen 50 MHz test frequency; Q (or E) per cycle is the frequency-independent comparison. Area is the gate-area proxy $\sum W \cdot L$ over all transistors (not a layout area).

Tools and flow. `run_all.py` (baseline, scaling sweeps, C_L and edge-rate sweeps, no-skew ablation); `optimize.py` (differential-evolution re-sizing at T_p targets 0.5 ns and 1.0 ns; constraints: $|T_p/\text{target} - 1| \leq 2\,\%$, $|t_{pLH} - t_{pHL}| \leq 15\,\%$ of T_p, output swing ≥ 1.7 V / ≤ 0.1 V, BBM node levels restored and isolation times > 0); `validate_opt.py` (re-runs the optimised sizes on the full testbench). Checks performed: (i) convergence errors (none in the final runs); (ii) no floating nodes (all nodes have DC paths; the DC operating point converged); (iii) independent Python trapezoidal integration of the raw waveform reproduces the ngspice charge to $< 0.01\,\%$ (67.349 fC vs 67.354 fC and 40.304 fC vs 40.310 fC); (iv) one bug turned up and was fixed: ngspice treats the name `gnd` as node 0, which silently shorted a current-sense source.

7. FUNCTIONAL VERIFICATION

The following are provided in this section:

- Input waveform
- Output waveform
- Truth table, where applicable
- Functional verification

Observation. Input and output waveforms of both circuits over one steady-state cycle (data from `conv_wave.csv` / `bbm_wave.csv` produced by ngspice) are in Fig. 3; the BBM internal skewed nodes are in Fig. 4. Both circuits are non-inverting overall (4 inversions).

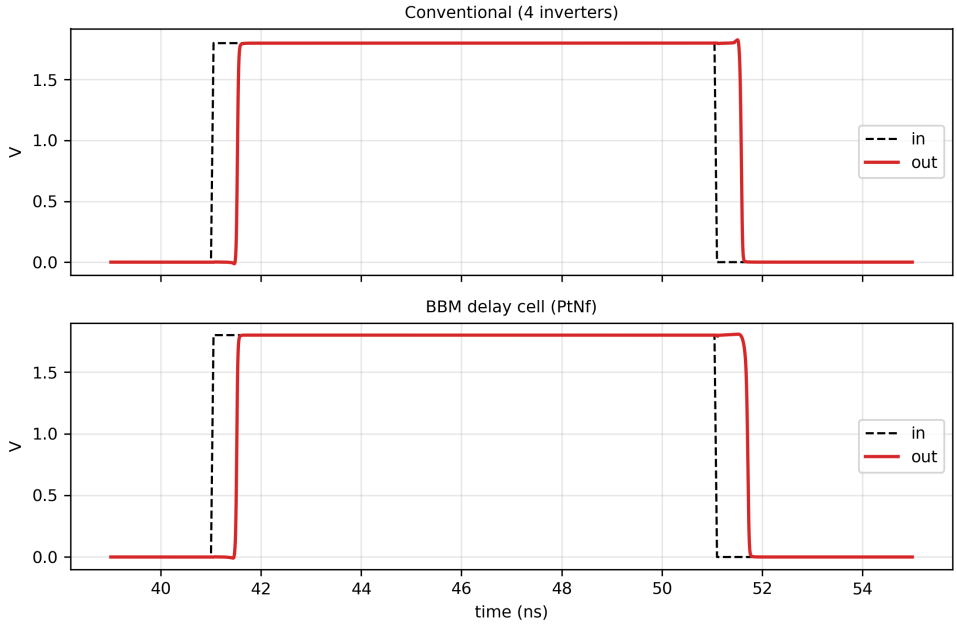


Figure 3. Functional verification: input (dashed) and output for the conventional and BBM cells (paper's 750 ps sizing).

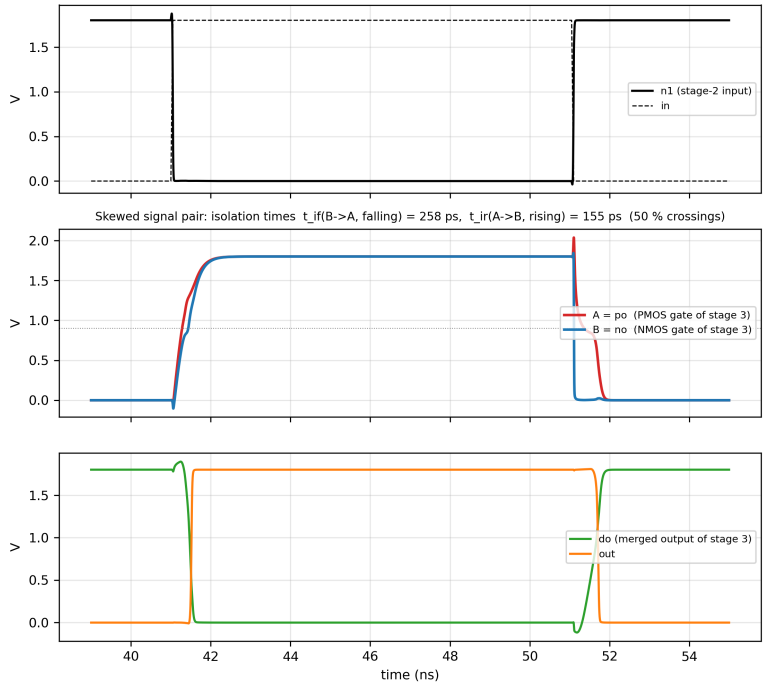


Figure 4. BBM mechanism: node A (PMOS gate of stage 3) and node B (NMOS gate of stage 3) switch at different times; the stage 3 output d_o follows.

in	out (ideal)	out (conventional, simulated steady state)	out (BBM, simulated steady state)
0	0	–0.014 V (logic 0)	–0.009 V (logic 0)
1 (1.8 V)	1	1.827 V (logic 1)	1.808 V (logic 1)

The table shows extreme values over the measured window, including small over/undershoot; settled levels are 0 V and 1.8 V.

Observation. Both circuits reproduce the input with a delay of about 0.5–0.6 ns and full-swing outputs ($V_{out,\max} = 1.83$ V / 1.81 V, $V_{out,\min} = -0.014$ V / –0.009 V for conventional / BBM). In the BBM cell, on a

rising input, node A (PMOS gate) rises 155 ps before node B (NMOS gate), and on a falling input B falls 258 ps before A (50 % crossings); an isolation interval therefore exists in every transition and the merge stage never has both gates in the conducting range at once. Node A overshoots to 2.04 V and node B undershoots to –0.10 V through gate-drain (Miller) coupling and then settle to the full rails (N_f restores the low level of A). Function verified.

8. PERFORMANCE MEASUREMENT

Column meaning: **Reference/Existing** is the conventional 4-inverter cell, re-sized (my own differential-evolution optimisation) for $T_p \approx 1.0$ ns. **Student Circuit** is my BBM cell exactly as built from the paper's published 750 ps sizing, not re-sized; its $T_p = 563$ ps does *not* match the reference, so it is shown for completeness only and must not be used for an improvement figure. **Optimized Circuit** is the BBM PtNf re-sized by the same optimiser for $T_p \approx 1.0$ ns. All values are from ngspice, full testbench, 27 °C, $C_L = 2$ fF.

Parameter	Reference/Existing	Student Circuit	Optimized Circuit
V_{DD}	1.8 V	1.8 V	1.8 V
Transistor Count	8	10	10
Power	5.84 µW	3.63 µW	4.48 µW
Delay	1016 ps	563 ps	1006 ps
PDP	5.93 fJ	2.04 fJ	4.51 fJ
Area	2.40 µm^2	1.00 µm^2	1.34 µm^2
Other	64.8 fC (116.7 fJ)	40.3 fC (72.5 fJ)	49.8 fC (89.6 fJ)

Supplementary comparison (conventional / BBM). The Δ columns give the change of BBM relative to conventional (negative means BBM is lower).

Case	T_p (ps)	Q (fC/cycle)	P_{avg} (μW)	PDP (fJ)	ΔQ	ΔPower	ΔDelay	PDP impr.
Paper 750 ps sizing (delays not matched)	507 / 563	67.3 / 40.3	6.06 / 3.63	3.07 / 2.04	–40.2 %	–40.2 %	11.2 %	33.5 %
Re-optimised, target 0.5 ns	495 / 505	44.1 / 38.3	3.97 / 3.45	1.96 / 1.74	–13.1 %	–13.1 %	2.0 %	11.4 %
Re-optimised, target 1.0 ns	1016 / 1006	64.8 / 49.8	5.84 / 4.48	5.93 / 4.51	–23.2 %	–23.2 %	–0.9 %	23.9 %

Optimised parameter sets (nm):

- 0.5 ns conventional: `lp1=779, ln1=1113, wp2=249, lp2=632, wn2=371, ln2=1257`
- 0.5 ns BBM: `lp1=913, wn1=239, ln1=196, wpt=260, lpt=189, lnf=283, lp2=197, ln2=593`
- 1.0 ns conventional: `lp1=1330, ln1=1223, wp2=283, lp2=1168, wn2=784, ln2=1426`
- 1.0 ns BBM: `lp1=1365, wn1=569, ln1=244, wpt=246, lpt=546, lnf=936, lp2=280, ln2=501`

9. OPTIMIZATION PERFORMED

Parameter Optimized: Transistor sizing of both cells (channel lengths and widths; for BBM mainly l_{P1}, w_{N1}, l_{N1}, w_{Pt}, l_{Pt}, l_{Nf}, l_{P2}, l_{N2} and the delay-element transistor P_t), for a fixed delay target, minimising the supply charge per cycle Q. Method: SciPy differential evolution driving ngspice (960 simulations per case).

Initial Value: Paper's 750 ps sizing: $Q = 67.3$ fC (conventional, $T_p = 507$ ps) and 40.3 fC (BBM, $T_p = 563$ ps).

Optimized Value: $T_p \approx 1.0\,\text{ns}$: $Q = 64.8\,\text{fC}$ (conventional) and 49.8 fC (BBM). $T_p \approx 0.5\,\text{ns}$: $Q = 44.1\,\text{fC}$ and 38.3 fC.

Reason: The paper's own comparison sizes every cell for the same delay. Comparing the BBM cell at the paper's sizing with an un-resized conventional cell ($Q = 67\,\text{fC}$ vs 40 fC, a misleading "40 %") is unfair, because the conventional chain at 750 ps sizing is far from the optimum for a 0.5 ns or 1.0 ns target (its charge falls to 44 fC at 0.5 ns). Re-sizing both circuits for equal delay gives numbers I can stand behind. Limits: the optimiser is much smaller than the paper's (100-node SADE), so neither sizing is guaranteed to be globally optimal.

10. SIMULATION RESULTS

Actual ngspice results (raw files in `results/`: `*.log` measurement logs, `*_wave.csv` waveforms, `summary.json`, `summary_opt.json`).

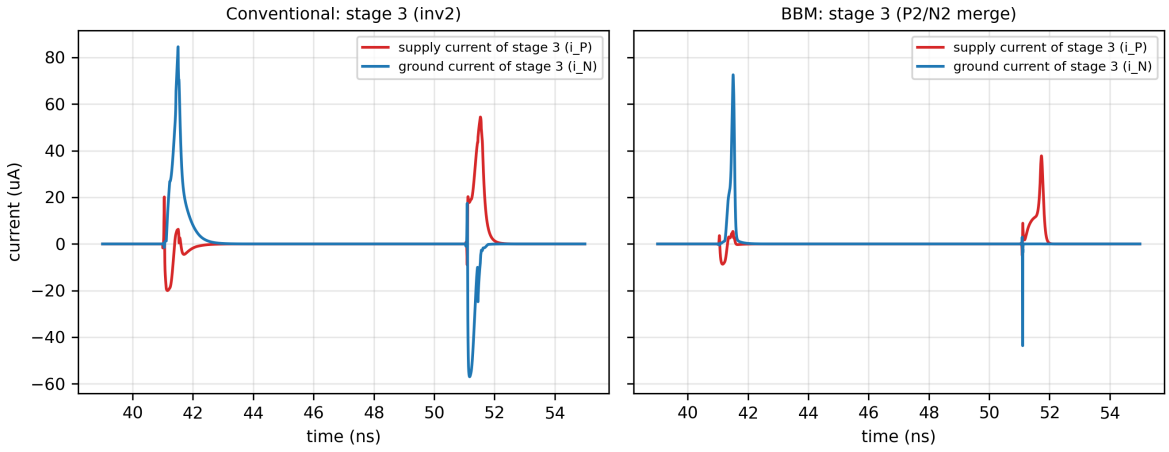


Figure 5. Stage-3 supply (i_P) and ground (i_N) currents. In the BBM cell the merged stage never carries a large overlapping current; the peak supply current of stage 3 is 37.8 µA (BBM) vs 54.4 µA (conventional).

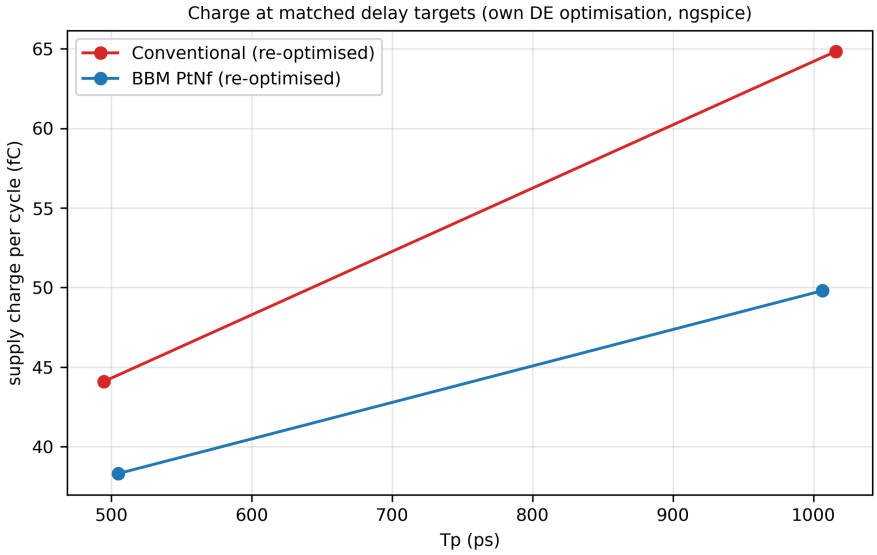


Figure 6. Charge per cycle at matched delay targets after re-optimisation (two targets only).

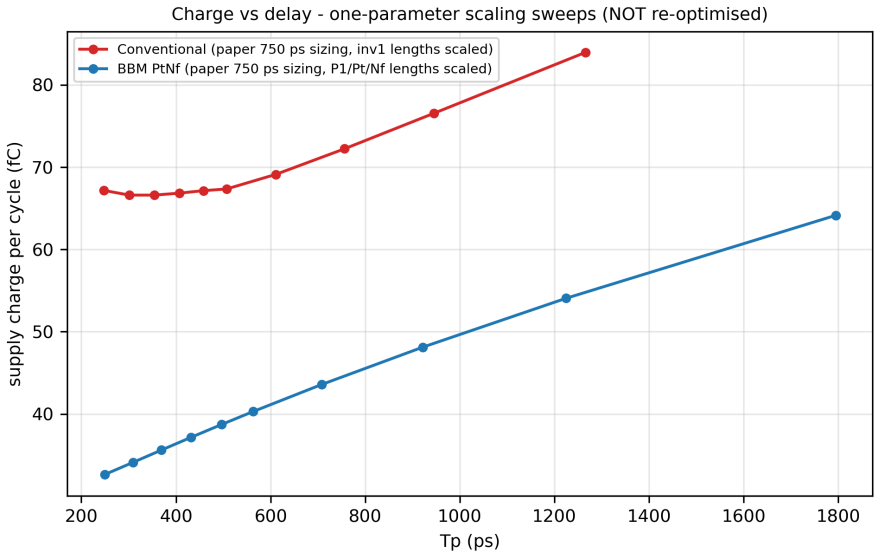


Figure 7. One-parameter scaling sweeps from the paper's 750 ps sizing (not re-optimised; the conventional cell is *not* at its optimum away from 750 ps, so this plot overstates the BBM advantage).

Load sensitivity at the paper's 750 ps sizing (delays not matched, so absolute savings are inflated; trend only).

C_L	T_p conv / BBM (ps)	Q conv / BBM (fC)	BBM charge change
0.5 fF	497 / 554	64.8 / 38.0	−41.4 %
2 fF	507 / 563	67.3 / 40.3	−40.2 %
5 fF	522 / 578	72.6 / 45.3	−37.6 %
10 fF	543 / 597	81.6 / 54.0	−33.9 %

Input-edge-rate sensitivity (same sizing).

Input edge	T_p conv / BBM (ps)	Q conv / BBM (fC)
20 ps	500 / 557	67.9 / 40.9
50 ps	507 / 563	67.3 / 40.3
200 ps	533 / 584	67.9 / 40.7
500 ps	562 / 606	70.5 / 43.2

Ablation (documented, not used for the improvement figure). Removing the delay element (A and B joined by 10 ■, N_f deleted) with identical device sizes gives $T_p = 155$ ps and $Q = 28.9$ fC, a different (much faster) circuit, because the P_t resistance is what creates the delay. It therefore cannot isolate the short-circuit saving. A simple $\min(i_P, i_N)$ crowbar estimate for stage 3 gives 0.74 fC (conventional) and 0.80 fC (BBM) per cycle, i.e. only a few percent of the stage-3 charge in this model.

Required Results:

- ✓ Functional waveform (Figs. 3, 4)
- ✓ Power measurement (Section 8 table)
- ✓ Delay measurement (Section 8 table)
- ✓ PDP calculation (Section 8 table)
- ✓ Optimization result (Section 9, Fig. 6)
- ✓ Comparative result (Section 8, Section 12)

11. RESULT

The implemented circuits were successfully simulated (ngspice-42) and the performance was evaluated. The improvement is computed for the delay-matched re-optimised pair at $T_p \approx 1.0\,\text{ns}$ (conventional 1016 ps, BBM 1006 ps).

Power: $P_{avg} = 5.84\,\mu\text{W}$ (conventional) vs 4.48 µW (BBM) at 50 MHz, 23.2 % lower ($Q = 64.8\,\text{fC}$ vs 49.8 fC per cycle).

Delay: 1016 ps (conventional) vs 1006 ps (BBM): 0.9 % shorter, i.e. matched by design (target 1.0 ns ±2 %).

PDP: 5.93 fJ (conventional) vs 4.51 fJ (BBM) (PDP $= P_{avg} \times T_p$ at 50 MHz).

Transistor Count: 8 (conventional) vs 10 (BBM).

Improvement Achieved: 23.9 % [$= (5.927 - 4.511)/5.927 \times 100$, from the logged PDP values].

12. DISCUSSION

At $T_p \approx 1.0\,\text{ns}$ the BBM cell cuts PDP by 23.9 % and charge by 23.2 %. At about 0.5 ns the cuts are 11.4 % (PDP) and 13.1 % (charge). The paper's Table 1 shows the same pattern: a 10 % saving at 0.5 ns, 36 % at 1.0 ns, a bit over 40 % from 3.5 ns, and no gain below about 0.35 ns. My saving at 1 ns is smaller than the paper's.

BBM is not free. It needs 10 transistors instead of 8, plus two wires between stages. At the paper's 750 ps sizing the BBM cell is also slower (563 ps against 507 ps, with rise and fall delays of 490 ps and 637 ps), so the 40 % charge saving at that point should not be quoted. The gate-area proxy at 1.0 ns is 1.34 µm^2 for BBM and 2.40 µm^2 for the conventional cell, but that is not a layout area. With sizes fixed, the gain shrinks as the load grows (41 % at 0.5 fF, 34 % at 10 fF) and with slow input edges (38.7 % at 500 ps). I did not test delays at or below 0.35 ns, where the paper finds no advantage.

Why does it save charge? Eq. (16) of the paper gives a charge proportional to T_p with a smaller slope than Eq. (9), because Q_s is about zero. In my runs the crowbar charge of the conventional stage 3 is small, so I think most of the saving comes from how the delay is made. BBM gets it from a long, narrow P_t (and long P_1 and N_f) that load only the small first stage. The conventional chain needs a big, long-channel inv2 whose capacitance is charged every cycle. The paper makes a similar remark about the third-stage gate capacitance. I did not separate the two effects rigorously.

Limits. Level-1 has no velocity saturation, sub-threshold current or gate leakage, so static power is zero, and only 27 °C was run. λ, γ and the capacitances are my assumptions, so absolute fC values cannot be compared with the paper's TSMC numbers. The unit-inverter sizes and C_L are assumed too. There is one corner, no mismatch, a reduced optimiser, no layout parasitics (as in the paper), and the input driver's charge is not counted.

13. CONCLUSION

I rebuilt the paper's break-before-make delay cell (PtNf) from its figures, wrote ngspice netlists for it and for a conventional 4-inverter reference, and simulated both with a substitute Level-1 0.18 µm-class model at 1.8 V. Both work and give full-swing outputs. In the BBM cell the PMOS-gate and NMOS-gate nodes of the last stage switch 155–258 ps apart (paper sizing), so the two transistors never conduct together.

With both cells re-sized for the same delay, BBM cut PDP by 23.9 % at about 1.0 ns and 11.4 % at about 0.5 ns, and the charge per cycle by 23.2 % and 13.1 %. That follows the paper's trend, more benefit at longer delay, but not its numbers. I would call it a qualitative confirmation for this substitute model, not a reproduction of the TSMC results.

14. REFERENCES

[1] J. Puhan, D. Raič, T. Tuma, and Á. Bűrmen, "Break-before-Make CMOS Inverter for Power-Efficient Delay Implementation," *The Scientific World Journal*, vol. 2014, Article ID 349131, 11 pp., 2014. doi:10.1155/2014/349131. https://pmc.ncbi.nlm.nih.gov/articles/PMC4265520/

[2] ngspice circuit simulator, version 42, https://ngspice.sourceforge.io/ (simulations and `.meas` measurements).

[3] Level-1 MOSFET model parameters taken from the caption of Fig. 4 of [1]. Unlisted parameters are my assumptions (see `models_180nm_L1.lib`).